

Exam Date & Time: 31-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
 BTech VII Semester - Make up Examination - December -2024
Social Network Analysis [CSE-4074]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate the density and cohesion of the given graph:

A)

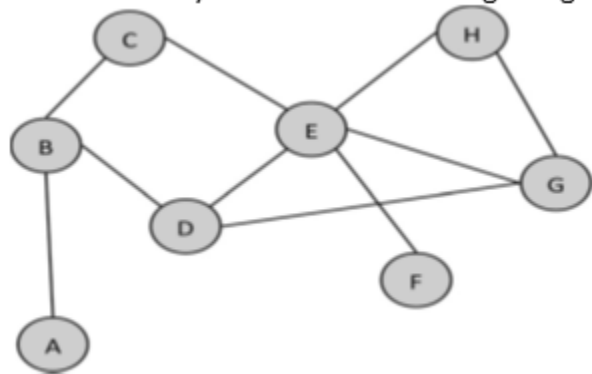


Figure 1: The Graph for reasoning

(5)

B)

Brief about Clique and clusters. Compute Clique and clusters for the following Figure 2:

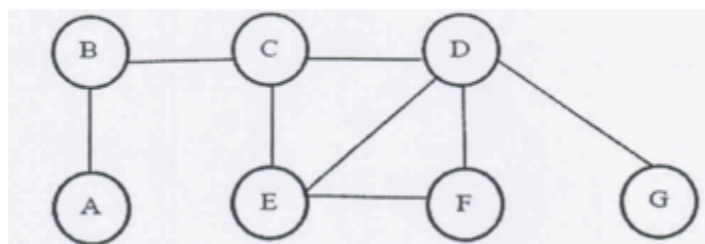


Figure 2: The Graph for reasoning

(5)

- 2) Calculate the density of each component in a network with three disconnected subgraphs:

A)

- Component 1: 4 nodes, 3 edges
- Component 2: 5 nodes, 5 edges
- Component 3: 6 nodes, 10 edges

(5)

B)

(5)

Illustrate the concept of cohesion? Find the cohesion of the following graph and explain in detail

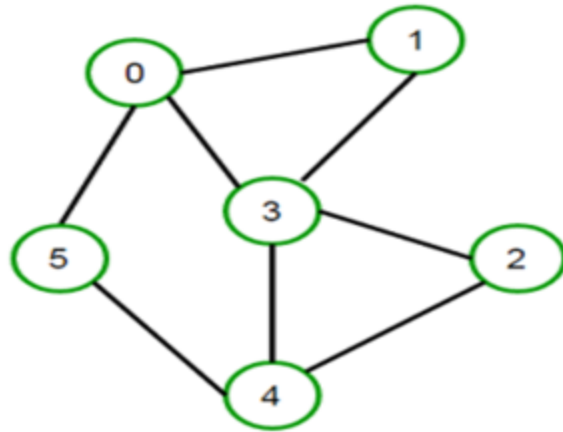
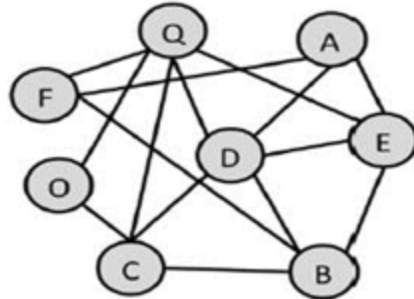


Figure 3: The Graph for reasoning

- 3) Calculate the scores and predict the most probable new edges using the Adamic-Adar method for the diagram provided below.

A)

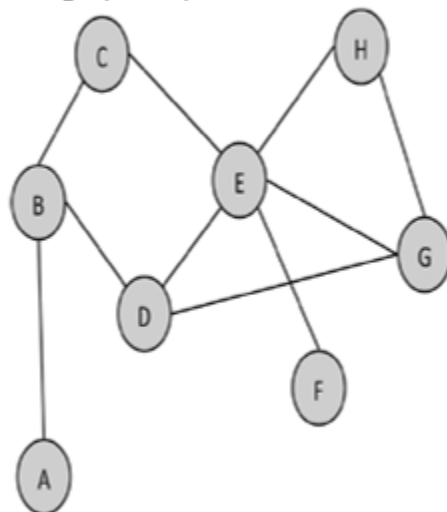


(5)

Figure 3: The Graph for reasoning

- B) Explain the common approaches used in link prediction?
- 4) Compute Laplacian Matrix and Normalized Laplacian matrix for this graph. Explain Newman fast algorithm.

A)



(5)

Figure 6: The Graph for reasoning

- B) You are analyzing a social network within a company to improve collaboration across different teams. During your analysis, you observe that certain employees serve as links between separate groups who otherwise have limited communication with each other. What is the concept of a structural hole in this scenario, and how do these linking individuals bridge these gaps? (5)
- 5) Examine the role of algorithms in influence maximization. Compare and contrast two algorithms used for influence maximization in social networks. (5)
- A)
- B) Describe the metrics used to identifying key influencers for statistics in social network analysis? (5)

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